HOMEWORK 2.2 · UNIT 2 · NC.8.EE.1

Exponent Rules (Multiplication & Division)

Aug 31 / Sep 1 · Targets: EE.1-2, EE.1-4, EE.1-1, EE.1-7, EE.1-5, EE.1-9, EE.1-8

Name _____ Date _____ Block _____

Remember: Same base, multiply → keep the base, **add** the exponents: $x^3 \cdot x^2 = x^5$. **Same base, divide** → keep the base, **subtract**: $\frac{x^7}{x^2} = x^5$. Monomials: multiply or divide the numbers, then use the rules on the letters: $(2x^3)(5x^4) = 10x^7$. $x = x^1$.

SCORE 3.0 Approaching · the basics

Targets: EE.1-2, EE.1-4, EE.1-1

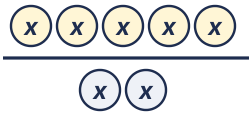
1 Simplify. Write as one power. $5^3 \cdot 5^4$ EE.1-2

2 Simplify. $(3a^2)(6a^5)$ EE.1-4

3 Simplify. $\frac{15y^6}{3y^2}$ EE.1-4

Ⓐ $5y^4$ Ⓑ $5y^3$ Ⓒ $12y^4$ Ⓓ $5y^8$

4 Use the picture. Cross out matching factors. Simplify $\frac{x^5}{x^2}$. EE.1-2



5 $2^5 = 10$ REVIEW 2.1 EE.1-1

True False

SCORE 4.0 Proficient · on target

Targets: EE.1-7, EE.1-5, EE.1-1

6 **Kim** made a mistake. Circle it. Explain it. Fix it. EE.1-7

KIM'S WORK
$x^4 \cdot x^3 = x^{12}$

7 Which are equal to 2^8? EE.1-5

- ☐ $2^4 \cdot 2^4$
- ☐ $2^2 \cdot 2^4$
- ☐ $2^{16} \div 2^2$
- ☐ $2^{10} \div 2^2$
- ☐ $2 \cdot 2^7$

Select all that apply.

8 Find the value. $(-2)^3$ REVIEW 2.1 EE.1-1

SCORE 5.0 Mastery · go deeper

Targets: EE.1-9, EE.1-8

9 Simplify $(4x^3y)(2xy^4)$. Tell what you did to the numbers and to each letter. EE.1-9

First I _____ the numbers. Then I _____ the exponents on x and on y. The answer is _____ .

10 Use the picture from #4 to explain why $\frac{x^5}{x^2} = x^3$. EE.1-8

Each x on the bottom _____ , so _____ factors are left.